

final_pds_revision

September 17, 2026

1 Data Science Lab Record - Master Revision Notebook

This notebook contains all the practical programs completed, organized by topic. It covers NumPy operations, Feature Engineering, Interactive Visualizations (Plotly & Bokeh), Web Scraping, and K-Means Clustering on the Iris dataset.

1.1 Table of Contents

1. NumPy Array Operations
 2. Feature Engineering & EDA
 3. Plotly Visualizations (Bar Chart)
 4. Bokeh Visualizations (Bar, Line, Scatter)
 5. Web Scraping & Data Collection
 6. K-Means Clustering on Iris Dataset (Raw Features & PCA)
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1.1.1 1. NumPy Array Operations

Problem Statement: Create a 2D array of shape (3, 4) with random integers between 0 and 10. Perform: mean, standard deviation, slicing (first 2 rows, last 2 columns), reshape to (4, 3), and element-wise multiplication.

1.1.2 2. Feature Engineering & Exploratory Data Analysis

Problem Statement: Create a DataFrame with random data, apply log transformation, bin a feature into 3 categories, calculate the correlation matrix, and visualize using a scatter plot colored by target.

1.1.3 3. Interactive Bar Chart with Plotly

Problem Statement: Create an interactive bar chart showing the population of different cities.

1.1.4 4. Interactive Visualizations with Bokeh

4.1 Interactive Bar Chart (Sales Figures) Problem Statement: Create an interactive bar chart showing sales figures of different products using Bokeh.

4.2 Interactive Line Plot (Daily Temperatures) Problem Statement: Create an interactive line plot showing daily temperatures over a week using Bokeh.

4.3 Interactive Scatter Plot (Iris Dataset) Problem Statement: Create an interactive scatter plot showing the relationship between petal length and petal width from the Iris dataset using Bokeh.

1.1.5 5. Web Scraping and Data Collection

Problem Statement: Scrape data from a webpage and store it in a structured format like CSV or JSON. Target: <https://quotes.toscrape.com>

Expected Output (quotes.csv):

Quote	Author	Tags
“The world as we have created it is a process...”	Albert Einstein	change, deep-thoughts, thinking, world
“It is our choices, Harry, that show what we...”	J.K. Rowling	abilities, choices
“There are only two ways to live your life...”	Albert Einstein	inspirational, life, live, miracle, miracles

1.1.6 6. K-Means Clustering on Iris Dataset

6.1 K-Means with Raw Features (3D Plot) Problem Statement: Perform K-Means clustering on the Iris dataset and visualize the clusters using an interactive 3D scatter plot in Plotly. Use raw features (no PCA).

6.2 K-Means with PCA (3D Plot) Problem Statement: Perform K-Means clustering on the Iris dataset and visualize the clusters using an interactive 3D scatter plot in Plotly. Reduce dimensions using PCA first.

```
[1]: # =====  
# IMPORTS  
# =====  
import numpy as np  
import pandas as pd  
import plotly.express as px  
import requests  
from bs4 import BeautifulSoup
```

```

from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from bokeh.plotting import figure, show, output_notebook
from bokeh.models import ColumnDataSource, HoverTool

# =====
# 1. NumPy Array Operations
# =====
print("--- 1. NumPy Array Operations ---")
# Create 2D array (3,4) with random ints 0-10
arr = np.random.randint(0, 11, size=(3, 4))
print("Original:\n", arr)

# Mean and Standard Deviation
print("Mean:", arr.mean(), "| Std Dev:", arr.std())

# Slice: first 2 rows, last 2 columns
print("Sliced:\n", arr[:2, -2:])

# Reshape to (4, 3)
reshaped = arr.reshape(4, 3)
print("Reshaped:\n", reshaped)

# Element-wise multiplication with same shape
arr2 = np.random.randint(1, 5, size=(4, 3))
print("Multiplied:\n", reshaped * arr2)

# =====
# 2. Feature Engineering & EDA
# =====
print("\n--- 2. Feature Engineering & EDA ---")
np.random.seed(42)
df = pd.DataFrame({
    'Feature1': np.random.uniform(1, 100, 150),
    'Feature2': np.random.uniform(1, 100, 150),
    'Target': np.random.choice(['A', 'B'], 150)
})

df['Log_Feature1'] = np.log(df['Feature1'])
df['Feature2_Category'] = pd.cut(df['Feature2'], bins=3,
    labels=['Low', 'Medium', 'High'])

print("Correlation Matrix:\n", df[['Feature1', 'Feature2', 'Log_Feature1']].
    corr())

```

```

px.scatter(df, x='Feature1', y='Feature2', color='Target',
           title='Feature1 vs Feature2').show()

# =====
# 3. Interactive Bar Chart with Plotly
# =====
print("\n--- 3. Interactive Bar Chart with Plotly ---")
data_pop = {'City': ['Mumbai', 'Delhi', 'Bengaluru', 'Hyderabad', 'Chennai', 'Kolkata'],
            'Population (Lakhs)': [204, 167, 136, 108, 115, 149]}
df_pop = pd.DataFrame(data_pop)

fig = px.bar(df_pop, x='City', y='Population (Lakhs)', color='City',
             title='Population of Different Cities', text='Population (Lakhs)')
fig.show()

# =====
# 4. Interactive Visualizations with Bokeh
# =====
print("\n--- 4. Interactive Visualizations with Bokeh ---")
output_notebook()

# 4.1 Interactive Bar Chart (Sales Figures)
data_sales = {'Product': ['Laptop', 'Smartphone', 'Headphones', 'Smart Watch', 'Tablet',
                           'Printer', 'Keyboard', 'Monitor', 'External Hard Drive', 'Webcam'],
              'Sales': [20000, 24000, 3750, 6000, 7000, 2250, 1800, 4500, 3500, 2000]}
df_sales = pd.DataFrame(data_sales)
source_sales = ColumnDataSource(df_sales)

p1 = figure(x_range=df_sales['Product'], title='Sales Figures of Different Products',
            x_axis_label='Products', y_axis_label='Total Sales ($)',
            height=450, width=850)
p1.vbar(x='Product', top='Sales', width=0.6, source=source_sales, color='steelblue')
p1.add_tools(HoverTool(tooltips=[("Product", "@Product"), ("Sales ($)", "@Sales")]))
show(p1)

```

```

# 4.2 Interactive Line Plot (Daily Temperatures)
data_temp = {'Day': □
    ↪ ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday'],
    'Temperature': [28, 30, 29, 31, 33, 32, 30]}
df_temp = pd.DataFrame(data_temp)
source_temp = ColumnDataSource(df_temp)

p2 = figure(x_range=df_temp['Day'], title='Daily Temperatures Over a Week',
    x_axis_label='Day', y_axis_label='Temperature (°C)',
    height=450, width=750)
p2.line(x='Day', y='Temperature', source=source_temp, line_width=2, □
    ↪ color='blue')
p2.scatter(x='Day', y='Temperature', source=source_temp, size=8, color='red') # □
    ↪ FIXED: p.circle deprecated
p2.add_tools(HoverTool(tooltips=[("Day", "@Day"), ("Temperature (°C)", □
    ↪ "@Temperature")]))
show(p2)

# 4.3 Interactive Scatter Plot (Iris Dataset)
iris = load_iris()
df_iris = pd.DataFrame(iris.data, columns=iris.feature_names)
source_iris = ColumnDataSource(df_iris)

p3 = figure(title='Relationship: Petal Length vs Petal Width (Iris)',
    x_axis_label='Petal Length (cm)', y_axis_label='Petal Width (cm)',
    height=450, width=700)
p3.scatter(x='petal length (cm)', y='petal width (cm)',
    source=source_iris, size=8, color='blue', alpha=0.6)
p3.add_tools(HoverTool(tooltips=[("Petal Length", "@{petal length (cm)}"),
    ("Petal Width", "@{petal width (cm)}")]))
show(p3)

# =====
# 5. Web Scraping and Data Collection
# =====
print("\n--- 5. Web Scraping and Data Collection ---")
url = "https://quotes.toscrape.com"
response = requests.get(url)
soup = BeautifulSoup(response.text, 'html.parser')

data_quotes = []
for q in soup.find_all('div', class_='quote'):
    data_quotes.append({
        'Quote': q.find('span', class_='text').text,
        'Author': q.find('small', class_='author').text,
    })

```

```

        'Tags': ", ".join([t.text for t in q.find_all('a', class_='tag')])
    })

df_quotes = pd.DataFrame(data_quotes)
df_quotes.to_csv('quotes.csv', index=False)
print(df_quotes.head())

# =====
# 6. K-Means Clustering on Iris Dataset
# =====
print("\n--- 6. K-Means Clustering on Iris Dataset ---")

# 6.1 K-Means with Raw Features (3D Plot)
X_scaled = StandardScaler().fit_transform(df_iris)
df_iris['cluster'] = KMeans(n_clusters=3, random_state=42, n_init=10).
    ↪fit_predict(X_scaled).astype(str)

px.scatter_3d(df_iris, x='petal length (cm)', y='petal width (cm)',
              z='sepal length (cm)', color='cluster',
              title='K-Means on Iris (3D Raw Features)').show()

# 6.2 K-Means with PCA (3D Plot)
pca = PCA(n_components=3)
df_iris[['PC1', 'PC2', 'PC3']] = pca.fit_transform(X_scaled)

px.scatter_3d(df_iris, x='PC1', y='PC2', z='PC3', color='cluster',
              title='K-Means on Iris (PCA)').show()

```

--- 1. NumPy Array Operations ---

Original:

```
[[8 2 3 5]
 [9 7 9 5]
 [2 6 3 7]]
```

Mean: 5.5 | Std Dev: 2.4664414311581235

Sliced:

```
[[3 5]
 [9 5]]
```

Reshaped:

```
[[8 2 3]
 [5 9 7]
 [9 5 2]
 [6 3 7]]
```

Multiplied:

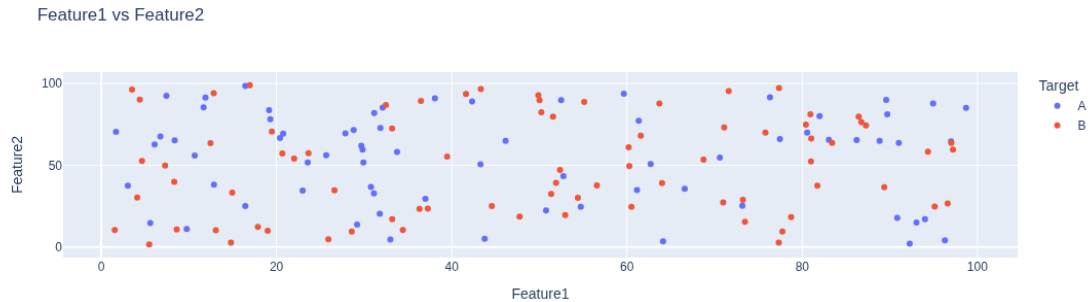
```
[[32  6  6]
 [15  9 28]
 [18 15  2]]
```

[12 6 28]]

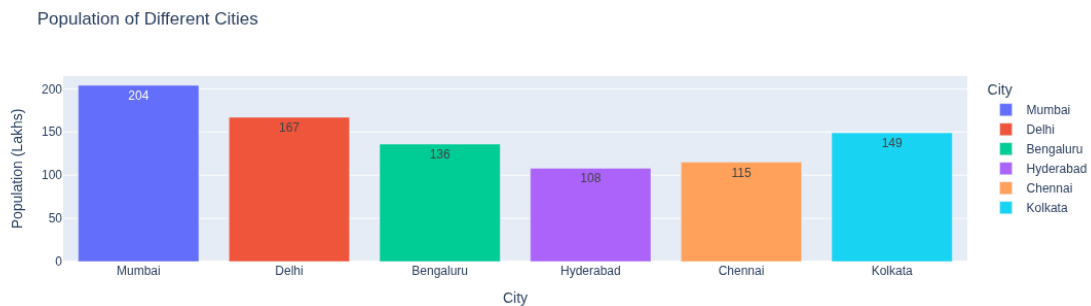
--- 2. Feature Engineering & EDA ---

Correlation Matrix:

	Feature1	Feature2	Log_Feature1
Feature1	1.000000	0.035691	0.909743
Feature2	0.035691	1.000000	0.040410
Log_Feature1	0.909743	0.040410	1.000000



--- 3. Interactive Bar Chart with Plotly ---



--- 4. Interactive Visualizations with Bokeh ---

--- 5. Web Scraping and Data Collection ---

	Quote	Author	\
0	"The world as we have created it is a process ...	Albert Einstein	
1	"It is our choices, Harry, that show what we t...	J.K. Rowling	
2	"There are only two ways to live your life. On...	Albert Einstein	
3	"The person, be it gentleman or lady, who has ...	Jane Austen	

4 "Imperfection is beauty, madness is genius and... Marilyn Monroe

Tags

0 change, deep-thoughts, thinking, world

1 abilities, choices

2 inspirational, life, live, miracle, miracles

3 aliteracy, books, classic, humor

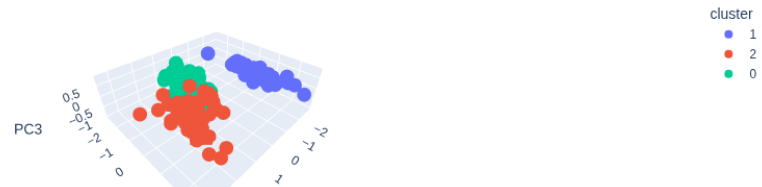
4 be-yourself, inspirational

--- 6. K-Means Clustering on Iris Dataset ---

K-Means on Iris (3D Raw Features)



K-Means on Iris (PCA)



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